ControlPlane.ai — Round 2 Final

Admission Control for AI That Acts

Accenture Innovation Challenge 2026 · Round 2 · dense hybrid (R2S1–R2S4)

ETERNAL: *Default = UNSUPPORTED · entitlement = set-membership (zero LLM) · exact R×S matrix · one graph · hard gate on actions · FNR as typed format · two-pending-actions (R1 Edit + R3 Escalate* ***held****, never "blocked") ·* `UNKNOWN` *never →* `SUPPORTED` *· bias async-only · refuse-to-claim (about us)*

Sources: `docs/ARCHITECTURE.md` · `docs/reference/NARRATIVE.md` · `docs/reference/QA.md` · `docs/ps.md` · *Stages 1–4 absorbed in this FINAL (locks deleted after merge).*

Official deliverable (`docs/ps.md`)	Artifact
Detailed Business Proposal	**This document**
Working Prototype	`controlplane/` · `examples/` · `tests/`
Pitch Presentation	R2S5.md + submission/ControlPlane_Round2_Pitch.pptx

Stage Check (compact)

Stage	Locked into	Status
R2S1 prototype scope	§§1–4	**PASS**
R2S2 enterprise envelope	§5	**PASS**
R2S3 build/demo	§6	**PASS**
R2S4 buyers/value/rollout	§§7–11	**PASS**

#	Eternal invariant	Status
1	Default = `UNSUPPORTED`	**PASS**
2	Entitlement = `span.acl` ⊆ `principal.clearance`; zero LLM	**PASS**
3	Exact R×S matrix; never redrawn; no route parameter	**PASS**
4	One graph: `STEP` → `SPAN` → `CLAIM` → `ACTION`	**PASS**
5	Hard gate on **actions**, not tokens	**PASS**
6	Dual-action: R1 **Edit** + R3 **Escalate** (**held**, never "blocked")	**PASS**
7	`UNKNOWN` never → `SUPPORTED`	**PASS**
8	FNR as typed format; empty until earned	**PASS**
9	Bias = async route-level only (never live matrix cell)	**PASS**
10	Refuse-to-claim list (about *us*)	**PASS**

1. Problem

Enterprises moved from AI that **answers** to AI that **acts**. Failure changed category: bad paragraph → **executed transaction**.

> *“Approved. Refund of INR 1,84,000 issued under clause 7.2 of the vendor agreement.”*

Clause 7.2 does not exist. Failure is *absence* of evidence, not conflict. Filters pass it. Confidence can read 0.94. Money moved Tuesday, found Friday. If ungated, **the company wrongly pays out INR 1,84,000** — the customer did not lose money.

This is not “hallucination” as the market frames it. It is an **authorisation** problem: an unproven claim authorized an irreversible action. Solution is not better text scoring — it is **admission control**.

> *The system didn’t fail. It was never asked to prove anything.*
>
> *Everyone watches the exit. Nobody records the entrance.*

Three structural reasons existing approaches fail:

1. **Inspect output, not context contract.** Without recording what the model was *given*, verification is unfalsifiable opinion. With it, verification is set-membership: *which span proves this claim?*

2. **Score the response, not the action.** Groundedness 0.82 means the same on a draft and a wire transfer. One threshold cannot price both.

3. **Identity-blind.** The common enterprise incident is a **correct answer to the wrong person**. No output-only inspector carries caller identity into verification.

Approach	Why it fails here
Post-hoc observability (LangSmith, Arize, …)	Traces *after* commit — audit trail, not interlock
LLM-as-judge / wrappers	Opinion, identity-blind, too slow for commit, cannot state own error rate
Static guardrails	Miss lexically clean fabrications and ACL failures
RAG groundedness	Averages (one wrong figure drowns); action-blind; **retrieval ≠ permission**
Confidence / logprob	Named failure is *confidently* wrong
Composite risk scores	Three owners/costs collapsed into one number that maps to no actuator

Failure modes this plane addresses: (1) categorical claim with **no provenance** authorizing R3; (2) claim bound to evidence the **caller may not read**; (3) **two pending actions, different blast radii** on one response; (4) no real-time ground truth → **invert burden of proof**.

Overlap (brief)	Treatment on one graph
Fabricated personal detail, no span	Unsupported + no-span PII/secret shape; stays UNSUPPORTED
Correct HR fact, ACL excludes caller	Entitlement violation — independent of semantic correctness
Absent policy clause authorizing refund	Unsupported categorical on R3 → Escalate (**held**)

Overlap (brief)	Treatment on one graph
Bias on decision-shaped outputs	**Not** a per-response verdict; async flip-rate + CI

Multi-turn = more STEPs on the **same session ledger** — not a separate product. Prior assistant text is never evidence by reappearance. Unproven turn-1 claim used to authorize R3 in turn N remains unproven. Tool results enter as SPANs, not automatic authorization. Compounding risk = same unproven claim reaching higher R — already covered by the frozen graph.

Enterprise question is not "how do we score AI responses?" It is: *What evidence is required before a claim may authorize a specific action, for this caller, on this route?*

Buyer question a sceptic can answer without our slide: *What consequential actions does this route perform today, what is the loss if one is wrong, and what fraction can sit behind an earned admission boundary?*

2. Solution

ControlPlane.ai is an **admission-control layer** (firewall / transaction validator class — not observability, not a second model). Deployed as thin context-assembly SDK hook + OpenAI-compatible reverse proxy. No weights, logits, or fine-tuning. Integration cost is real — **and is the moat**.

```
AI output → claim proof → entitlement → R×S → action admitted / edited / escalated / held
```

One primitive, three reads:

```
STEP ■■produces■■■ SPAN ■■binds■■■ CLAIM ■■authorizes■■■ ACTION
```

Performance reads forward. Cost reads backward (exact dead compute). Responsibility reads labels. One structure — not three detectors.

Load-bearing primitives

1. **Provenance outside the model** at context assembly — keystone: `source_id · ACL · content_hash · offsets` + calling principal. Model has **no write path**. Binding computed by the plane; model-emitted citations are not evidence; no open-web rescue.

2. **Default = UNSUPPORTED.** Earn `SUPPORTED` via recomputation or binding. Unsupported ≠ low confidence — **unproven**. `UNKNOWN` never → `SUPPORTED`.

3. **Claim-type routing.** Numeric/date/ID → recompute. Textual → bind (entailment, not string match). Derived/multi-hop → recompute or `UNKNOWN`.

4. **Entitlement = set-membership** (`CALLER → CLAIM → SPAN → SOURCE ACL`; `span.acl ⊆ principal.clearance`). Zero LLM. Lane 1 always on. Semantically correct + unauthorized still fails. **Most differentiated mechanism** — competitors cannot replicate without carrying identity into verification.

5. **Exact R×S matrix per pending action.** Disposition = **worst claim for that action — never an average**. `R = irreversibility × audience × data class × autonomy`. Action Interlock = sole final decider; pure rule engine.

6. **Hard gate on actions, not tokens.** Text hold-back ~150–300 ms. Speculative verification OK; **speculative release forbidden**.

7. **Escalate → evidence packet** (claim, spans, verdict, diff). Surgical Edit only (strip or one constrained regen; re-gate; second fail → Escalate).

8. **Publish own per-route FNR as typed format** — empty until earned. Emptiness is the credibility play.

Ungrounded / parametric: when no provenance spans exist for a route, it is declared ungrounded by construction; such claims **cannot authorize any action** regardless of blast radius. *We don't claim to verify what we were never given. We claim what we were never given cannot authorize an action.*

Refuse-to-claim (about *us*, not competitors)

1. **"We eliminate hallucinations."** False if shipped. Narrower claim: ungrounded claims cannot authorise actions, and we report what we miss.

2. **"Zero integration, drop it in."** We hook context assembly — real work, exact reason the design works. Scope = one SDK hook + OpenAI-compatible proxy (days on a standard retrieval stack, not quarters) — never sold as drop-in. Integration cost is the moat.

3. **"Zero added latency."** We never make the model feel slow; we make the action wait. Quote only ≤**40 ms p50 /** ≤**200 ms p95** — never 40 as p95.

4. **"One accuracy number across failure modes."** Hallucination, leakage, and bias have different mathematics, error costs, and owners.

Operating claim: ControlPlane does not promise "safe" or "truthful" AI. It makes an unproven or unauthorized claim unable to authorize an action beyond the route's admitted boundary — and publishes what the plane missed.

3. Matrix

Exact frozen 16-cell R×S — transcribed, never redrawn, **no route parameter**:

	Contradicted / entitlement violation	Unsupported + categorical	Unsupported + hedged	Unknown
R3	**Block**	**Escalate**	**Escalate**	**Escalate**
R2	**Block**	**Edit**	**Edit**	**Escalate**
R1	**Edit**	**Edit**	**Pass + annotate**	**Pass + annotate**
R0	**Pass + annotate**	**Pass + annotate**	**Pass**	**Pass**

R tiers: R0 = internal draft · R1 = user-visible read-only · R2 = reversible write / external send · R3 = irreversible or regulated (payment, deletion, publication, regulated advice).

Actuators (exact): Block · Edit · Escalate · Pass / Pass + annotate. Forbidden inventions: `STREAM`, `Kill Span`, `Hold & Re-verify`, `Redact & Flag`, `COMMIT BLOCKED` as R3 unsupported-categorical label. Autonomy downgrade / circuit breaker = spoken controls, not inventable demo labels.

Fail stance ≠ matrix: Matrix answers "evidence arrived and is bad → actuator?" Fail stance answers "evidence did not arrive in time?" Floors: R0/R1 fail **open + annotate**; R2/R3 fail **closed or escalate**. Universal fail-open forbidden.

Centrepiece cells (simultaneous on one refund response):

Pending action	Tier	Finding	Cell	Actuator
Show text	**R1**	Unentitled span grounds a claim (C3)	R1 × entitlement	**Edit**
Issue refund	**R3**	Clause 7.2 has no span (C2)	R3 × unsupported-categorical	**Escalate — held** with packet

Never collapse into one response-level verdict. **Never say “blocked” about the refund.** Same unsupported+categorical claim yields R1→Edit on a draft and R3→Escalate on a payment — proof scales with consequence; the matrix is not renamed severity.

4. Prototype

Exactly **two** live routes. Third live decision-support / bias route **refused**. Bias = proposal measurement only. Scope tension (resolved): brief names decision-support and bias; architecture defines bias as distributional/async — both stay in the **enterprise envelope**, never contaminate the live core.

Route	Dominant R	Mechanism proved	Live?
A. Refund agent	R1 text + R3 refund	Two-pending-actions	**YES**
B. Knowledge assistant	R0/R1	Provenance + entitlement flip	**YES**
Decision-support / bias	—	Async counterfactual	**NO** — envelope only

Refund risk signature: absent clause · numeric/ID error · entitlement leak into customer text · irreversible payment on same response · no real-time ground truth. Latency: R0/R1 text ≤**40 ms p50 / ≤200 ms p95** (never quote 40 as p95); action verify amortised in tool RTT; text hold-back ~150–300 ms. Speculative verification OK; speculative release forbidden.

Knowledge: correct-but-unauthorized disclosure; fabrication; ACL-excluded span. Demo stages ACL-excluded → R1 × entitlement = Edit (not no-span PII → Block). Plane enforces carried ACLs — does **not** repair IAM. Authorization path: `CALLER → CLAIM → SPAN → SOURCE ACL` — set-membership, **zero LLM**.

Why this pair: Refund proves one graph, two actuators, hard gate, surgical edit, packet. Knowledge proves graph ≠ RAG groundedness: same claim, different principal → different outcome (RAG averages; ControlPlane prices worst claim per pending action). Together: *is this claim proven?* and *is this caller entitled to that proof?* A third live bias route adds async measurement without strengthening the dual-action centrepiece.

Assumptions (load-bearing): synthetic enterprise-shaped corpora only — no real PII; every span carries source/ACL/hash/offsets; generator via OpenAI-compatible API only (no weights/logits); decision policy = pure rule engine; traffic directional tens of thousands/week (prototype = curated single-node — no throughput claim); FNR null without trustworthy ground truth; corpus poisoning outside truth guarantee (hash makes source auditable, not factually true); DPDPA/GDPR as config packs — not certification.

Boundary — in

Prove	How
Provenance outside model	Spans before claims; model cannot author
One typed ledger	`STEP → SPAN → CLAIM → ACTION`
Default UNSUPPORTED	Earn via bind/recompute; UNKNOWN never → SUPPORTED
Claim-type routing	Numeric recompute; textual entailment; derived → UNKNOWN
Deterministic entitlement	Set-membership; zero LLM
Exact 16-cell matrix	Per pending action; worst-claim weighting
Dual-action centrepiece	R1 Edit + R3 Escalate held

Prove	How
Hard action gate	Mock refund `committed:false` while Escalate
Surgical Edit + evidence packet	Strip/regen once; packet = claim+spans+verdict+diff
Empty FNR schema	Typed nulls only — no fabricated %
Principal flip	Change only caller → outcome flips
Per-claim surface	Verified / Uncertain / Blocked; refund action = Held/Escalate
Ungrounded / parametric	No provenance → cannot authorize any action
Live compute	Binding/entitlement/interlock visible (~20–80 ms)

Boundary — out

Third live bias route · per-response bias · production load as mechanism proof · real payments/PII · live IAM repair · LLM-as-judge primary · confidence/risk scores · open-web truth · generative rewrite · fabricated FNR · Lane-3 on critical path · triage SLA UI · weights/logits · collapsing dual-action · calling refund “blocked.”

Measured results (this build)

All numbers are produced by `make eval` and `make bench` and regenerate on every run — they are measured, never asserted. The capability ledger and per-route Wilson CIs live in the generated README PDF (`submission/ControlPlane_Round2_README.pdf`) and `evals/last_run.json`.

This build earns the FNR on a 168-case self-authored corpus. The committed run (`evals/last_run.json`) reports:

bullet **Ungrounded FNR 1.1%** (95% Wilson CI 0.2%–5.8%; n=94 ungrounded claims; 1 published miss).

We publish the interval — the honest claim is the interval, not a fake-perfect 0% and not a tighter bound than the data supports.

bullet **Passable-action FPR 0.0%** (95% Wilson upper bound 15.5%; n=21 passable claims).

bullet **Hard-negative hold rate 64% [50.7%, 75.7%].** 53/168 cases are hard negatives (31.5%, above the 20% floor). We over-flag — that is our named next milestone. Clean strata hold 0/13.

bullet **One published miss:** `struct-miss-000` — a response citing "Clause 4.1 permits this refund" where the span only says clause 4.1 *covers shipping delays*. Structural symbol lookup matches → SUPPORTED → low-tier action slips through. We show the judge exactly which case we got wrong.

bullet **No production proof.** Production FNR is unknown until shadow replay over live traffic (ARCHITECTURE §7, §12). The honest claim: "on this self-authored corpus we miss 1.1% of ungrounded claims (95% Wilson CI 0.2–5.8%; n=94); production is unknown."

bullet **Refuse-to-claim:** we do not eliminate hallucinations; we do not claim drop-in integration; we do not claim zero added latency; we do not claim one accuracy number; we do not fill FNR with fabricated percentages.

Shape of the published claim (ARCHITECTURE §7): we hold **X%** of ungrounded responses at **Y ms** p50 — and here is the **Z%** we don't, with a Wilson interval.

5. Enterprise Envelope

Same plane on every route. Runtime:

```
App/Agent → Context SDK Hook (SPAN: source·ACL·hash·offsets)
         → OpenAI-compatible proxy
         → Evidence Ledger STEP→SPAN→CLAIM→ACTION
         → Action Interlock  exact R×S = f(R,S)  [no route param]
         → Pass / Pass+annotate / Edit / Escalate / Block
         → Text hold-back · Action commit boundary
```

Shared (not soft-configurable): Provenance Recorder · Ledger · typed claims · proof paths · verdict set · Entitlement Auditor · Interlock · exact matrix · fail-stance floors · hard gate · surgical edit · packet · claim-level surface · FNR schema · Lane 1 always on.

RoutePolicy one-liner (configures budget, not truth):

```
RoutePolicy { route_id · tenant · use_case · provenance_scope · action_grammar
  · action_to_R_mapping (locked R3: payment|deletion|publication|regulated_advice)
  · verification_profile · fail_stance_by_R (may NOT loosen R2/R3 to fail-open)
  · enforcement_mode shadow|canary|enforce · error_budget · escalation_target
  · sampling_policy · geography_overlay (additive only) · latency_budget ≤40/≤200 }
```

Routes may change actions, R-mapping (except locked classes), Lane-2 budget, shadow/enforce, sampling, additive geo. Routes may **not** change: UNSUPPORTED default · entitlement · matrix cells · hard gate location · UNKNOWN→SUPPORTED · composite scores · Stage 1 dual-action scope. Low-consequence traffic gets **less verification budget and a proportionate actuator — not weaker truth semantics**.

Route profile	Dominant R	Lane posture	Enforcement
Customer-support refund	R1+R3 simultaneous	Lane 1 text; Lane 1+2 + interlock for R3	Shadow/canary before enforce; **demo still shows hard gate on mock refund**
Internal knowledge	R0/R1	Lane 1 dominant; ACL always inline	Enforce after shadow evidence
Decision-support	R0/R1 memo + R2/R3 action	Lane 2 derived; Lane 3 async bias	Action gate when earned; bias always async — **not Stage 1 live**

Lane	Runs	Path
1	Span membership, ACL, arithmetic, typed interlocks, PII shapes	Inline; always on; no LLM
2	NLI/binding for flagged textual / high-R	Near-line; timeout → UNKNOWN
3	Semantic-entropy, bias replay, calibration, shadow audit	Async; **never** critical path

Governance: versioned DAG of `(signal, threshold, action, latency_budget)`; decision time = pure rule engine. Hierarchy narrows only: Enterprise baseline → geo overlay (additive) → tenant → route → action class. Overlays cannot loosen frozen invariants. Global parse-time invariants include: default UNSUPPORTED · UNKNOWN→SUPPORTED forbidden · entitlement always on · Lane 1 cannot disable · locked R3 classes · matrix immutable · R2/R3 fail closed/escalate · hard gate on actions · no composite score · no LLM at decision · bias never a matrix verdict · Stage 1 scope unreopened.

Policy lifecycle: Draft (hashed) → static validation → shadow replay → canary → auto-rollback if override >3× baseline or error budget breached → named-principal approval → promote. Day-one = **shadow**. Production enforce is **earned**.

Audit: `principal · evidence · claim verdict · matrix cell · actuator · policy_version · verifier_versions · latency · lane · route_id` — reconstructible from ledger + policy + frozen matrix.

Feedback = calibration, not adaptation. Learn thresholds/error-budget candidates via shadow→canary. Never learn: confidence/logprobs · silence-as-safety · aggregate override as permission to loosen · poisoned sources · prior assistant text as evidence · LLM-as-judge opinions · online weight updates. Override = evaluation evidence, not real-time security rewrite.

FNR schema (null until earned):

```
route_id · policy_version · evaluation_window · strata_definitions
· sampled_count_per_stratum · false_negative_count · ground_truth_positive_count
· FNR_estimate · CI_lower · CI_upper · ground_truth_method
· measurement_status  # null|insufficient_sample|prototype_corpus|production_measured|stale
· limitations
```

Method: stratified shadow audit — 100% of Block/Escalate/Edit + sample of Pass; ground truth = human / expensive multi-verifier — **never** LLM-as-judge. Claim shape when earned: *"On this route we catch \<measured\>% of ungrounded claims at 40 ms p50 — and here is the \<measured\>% we don't."*

Bias (brief requirement, frozen stance): async route-level counterfactual flip-rate + CI; flag when CI excludes zero. **Never** `claim → bias verdict → matrix`. Never dropped; never a Stage 1 live route.

Layer	Enterprise	Stage 1 live	Proposal-only
Routes	Support + knowledge + decision + agentic	**Exactly two**	Extra packs
Core	Full plane	Dual-action + flip + ledger + empty FNR	Multi-tenant HA
Bias / FNR values	Async + measured when earned	Absent / null placeholders	Full programs
Scale	Tens of thousands/week (directional)	Curated single-node traces	Load/HA

Prototype = proof the plane works. Envelope = proof the plane can be operated. Pitch = both. Envelope residual risks (architecture): (1) derived/multi-hop false SUPPORTED → recompute-or-UNKNOWN + stratified FNR; (2) poisoned sources / wrong ACLs → hash + entitlement-by-source detector — defend claim↔evidence link, not source truth; (3) over/under-flag + R mis-mapping → shadow earn-out, locked R3, interlock in executor. Stage 2 IS configuration/governance/lifecycle around the primitive. Stage 2 IS NOT a second mechanism, redrawn matrix, live bias product, or trained safety model.

6. Build & Demo

Corpora (essentials — no breadth)

Source	ACL	Purpose
`AGR-VENDOR-v3`	agent-readable	Clauses 1–6 only; **no clause 7.2 anywhere**

Source	ACL	Purpose
`ORD-1023`	`refund_agent`	amount=184000 INR — C1 SUPPORTED
`FIN-INTERNAL-NOTE`	`internal_analyst` (excludes `agent_refund_7`)	C3 entitlement → Edit
`INJECT-NOTICE`	untrusted	Cannot author provenance
`HR-COMP-L6`	`hr_partner`	Principal-flip span

Principals: `agent_refund_7 · analyst_01 · hr_partner_01`. Canonical fixture: *"Approved. Refund of INR 1,84,000 issued under clause 7.2 of the vendor agreement."* + one FIN-INTERNAL sentence. Span contract: `span_id · source_id · ACL · content_hash · offsets · text · step_id`.

Claim	Finding	Feeds
C1 amount/order numeric	Binds `ORD-1023` → **SUPPORTED**	Proof works
C2 "under clause 7.2 …" categorical	**No span** → stays **UNSUPPORTED** (absence ≠ contradiction)	`refund.execute` → R3 **Escalate**
C3 grounded on FIN-INTERNAL	ACL excludes caller → **entitlement**	`text.show` → R1 **Edit**

All three born UNSUPPORTED. Optional polish only after dual-action+flip green: paraphrase entailment · parametric ungrounded · numeric near-miss — cut polish before cutting dual-action.

Components

Real: Recorder · Ledger · Claim Extractor (fixture) · Numeric Recomputer · Entitlement Auditor · Interlock · Surgical Editor · Packet · Hold-back · Policy loader · Principal switch · FNR renderer · Trace Console · harness. Thin mock: generator stub · refund executor (`REFUND HELD` / `committed:false` — never `COMMIT BLOCKED`) · NLI labels. No LLM at decision time. No confidence field in the decision path.

Demo order (≤8 min; UI ≥60% ledger)

Governing test: if removing the graph leaves the demo looking the same, scope failed. Open on held transaction — never a risk statement. Core crisis ≤90s. First spoken sentence about failure contains **claim**, not response.

Must show	Must NOT show
Ledger ≥60%; spans before claims; matrix cells before actuators	Composite risk/confidence scores
Action Gate cold-open with `committed` boolean	"Response blocked" / `COMMIT BLOCKED` for R3 unsupported-categorical
Per-claim Verified/Uncertain/Blocked; refund = Held/Escalate	LLM-as-judge pane · open-web lookup
Evidence packet on every Escalate; empty FNR	Interlock bypass override · third-route chrome · bias widget · chatbot-majority

1. **Cold open gate:** `refund.execute {amount:184000, reason:"clause 7.2", order_id:"ORD-1023"}` · R3 · **HELD — ESCALATE** · `executed:false`
2. Expand ledger — spans with source·ACL·hash **before** claim verdicts
3. C1 SUPPORTED · C2 no span UNSUPPORTED · C3 entitlement violation

4. Highlight cells **before** actuators: `text.show` → R1×entitlement→**Edit**; `refund.execute` → R3×unsupported-categorical→**Escalate**

5. Surgical Edit strips C3; refund stays held; packet opens for C2 (claim, candidate spans `[]`, verdict, diff)

6. Executor proves not committed — company does **not** wrongly pay out in the gated demo

7. Empty FNR schema visible (null placeholders only)

8. **Principal flip (≤2 min):** `analyst_01` → Edit; flip only to `hr_partner_01` → Pass. Zero LLM.

Optional third beat (cut first): same unsupported claim as R1 vs R3 → actuator changes solely because R changed — or numeric `14` vs span `11` → R1 Edit.

Voice: authorise · admit · prove · bind · refuse · hold · escalate · gate — not monitor · detect · observe · watch · guard · trust score · risk score.

Run

```
python3 -m venv .venv && source .venv/bin/activate
pip install -e ".[dev]"
python3 -m pytest tests/ -v
python3 examples/refund_trace_demo.py
python3 examples/knowledge_flip_demo.py
```

Command	Proves
`pytest tests/ -v`	Matrix, entitlement, UNSUPPORTED default, hard gate, held≠blocked, R2S1 criteria on fixtures
`refund_trace_demo.py`	Dual-action: Edit + Escalate held; never collapsed; never “blocked”
`knowledge_flip_demo.py`	Same span/claim; outcome flips with clearance only

7. Buyers

Target users

Beachhead = high-consequence routes (refund-class R3 + mixed-governance knowledge) — not “all enterprise AI safety.”

Role	Who	Buys / operates
Economic buyer	Ops / CS / CHRO / CRO / CFO / CISO / Risk	Action authorisation + per-route error budgets. Responds to: *unproven claim cannot authorise an action.* Not: *we detect 95% of hallucinations.*
Technical buyer	Platform / ML infra / Identity	SDK hook + proxy + adapters; wants graph, not second black box; ≤40/≤200 ms
App / agent team	Route owners	Wire adapters; no app rewrite

Role	Who	Buys / operates
Day-to-day actor	Support / knowledge worker	Sees Verified/Uncertain/Blocked per claim; Held/Escalate on action — never enforces
Operator / governor	Route owners, SRE, Risk Ops	Shadow/canary/enforce; rollback; circuit breaker; ledger reconstruct
Escalation reviewer	Risk / ops	Evidence packet, not bare alert
Compliance / audit	Legal / DPO / audit	Influences via reconstructible trail

Split that matters: who **pays when it fails** ≠ who **runs it** ≠ who **types the answer**. Interlock enforces; plane publishes miss rate.

8. Impact Logic

Business case

No fabricated ROI. No "99%." No net-savings slide. Value = **mechanism** → **consequence**; buyer fills middle terms. Tail risk, not average case.

```
Exposure = freq(consequential AI actions) × P(unproven/unauthorized claim) × loss/wrong action
```

Lever	Mechanism → consequence
A — Avoided wrong actions (primary)	R3 Escalate held (`executed:false`) is structural escape. Value = held true-positives × buyer's direct cost of that action class; residual sized by published FNR. Artifact: attempted commit · matrix cell · gap · `committed=false`.
B — Blast-radius pricing	Lane 1 majority volume; expensive proof where R justifies. Same verdict annotates a draft and holds a payment. Answers brief's one-size-fits-all latency problem.
C — Exact dead compute	Backward walk: STEP grounding zero accepted claims = waste (exact). Observability measures spend; this measures waste. Expose number; enterprise prices own traffic — no % saved on slide.
D — Alert fatigue without lowering gate	Matrix prices actuators; R0/R1 unsupported+hedged → Pass+annotate. Shadow earn-out. Metrics: overrides · gate-fail · edit/escalation **per route**.
E — Auditability	Hash-chained ledger + versioned policy. Reconstruct: action → cell → verdict → span → source/hash/ACL → principal → policy → latency. Regulator answer = pointer.
E2 — Unauthorized disclosure	ACL-excluded span caught when semantically correct. Does not fix IAM — stops IAM gaps being **silently bypassed by a model**.

Lever	Mechanism → consequence
F — Publish misses	Per-route FNR typed format; null until earned. Conversation: route · population · miss · CI — not “trust our safety score.”
G — Earned autonomy (secondary)	Shadow → enforce only where counterfactuals justify. More AI action **after** evidence — never lead claim.

9. Roadmap

Earn-out, not feature calendar. Enforcement earned, not switched on. Day-one = shadow. Misses published every phase.

Phase	What	Exit
0 — Prototype	Two live routes; synthetic corpora; ledger ≥60%; empty FNR	All R2S1/R2S3 criteria; dual-action crisis ≤90s
1 — Shadow production	1–2 real routes matching pattern; dual-emit; no production hold yet. Determinism works day one; statistics earn thresholds	Counterfactuals enough to open canary
2 — Canary R0/R1	Full policy lifecycle; enforce Edit/Pass+annotate on read-only	Tolerable ops; no material edit regression
3 — Limited R2/R3 enforce	Hard-gate selected actions; packets to humans; fill FNR only with trustworthy GT; locked R3 at parse	Buyer FNR/FP/override thresholds; `committed=false` on held TPs
4 — Broader envelope	More routes (decision-support **template**); additive geo; async bias; dead-compute FinOps; circuit breaker; fail stance tier-owned	Multi-route without second detector

10. Risks

Risk	Why it matters	Mitigation
False assurance on derived/multi-hop	Shallow entailment → false SUPPORTED — worse than no plane	Bypass NLI → recompute or UNKNOWN; UNKNOWN never SUPPORTED; timeout→matrix; FNR by claim type
Poisoned sources / wrong ACLs	Plane proves claim↔evidence + carried ACLs — not source truth / not IAM repair	Immutable source+hash; missing ACL→unentitled; entitlement-violation **by source**; quarantine via policy
Prompt injection / model-declared bindings	Invent evidence edges	Plane computes bindings; model cannot author spans; disposition = rule engine

Risk	Why it matters	Mitigation
Over-flag → bypass; under-flag → liability	Classic guardrail death	Shadow default; blast-radius pricing; fail-stance floors; override>3× rollback; circuit breaker
R mis-mapping (payment as R1)	Wrong row without “breaking” matrix	Locked R3 at parse; interlock in **executor**, not only UI
Integration / provenance gaps	Incomplete spans → false UNSUPPORTED	Honest hook+proxy; evidence coverage measured; conservative high-R
Plane as SPOF	Outage / load-induced bypass	Tier-owned fail stance; universal fail-open forbidden
Feedback misuse / weight access	Train on overrides; require logits	API-only; feedback→policy candidates only; calibration ≠ trained judge
Switch-off after a quarter	Layers that interrupt low-blast text die	Gate on actions; majority Pass+annotate; earn-out before trust; integration cost stated as moat
Pattern-match as RAG/safety dashboard	Differentiation dismissed	Open on held INR 1,84,000; ledger majority UI; ban monitor/detect/risk-score vocabulary

11. Differentiation

ControlPlane	Everyone else
Provenance outside model at context assembly	Inspect output; treat model citations as evidence
Default = UNSUPPORTED	Default allow; flag what looks wrong
Entitlement = ACL set-membership	Identity-blind scorers
One graph, three reads	Three bolted tools
Exact R×S per pending action	Composite risk / confidence
Hard gate on actions	Gate on text/tokens
Publish per-route FNR (misses)	Publish precision (bother rate)

Ordered contrast

1. **vs observability** — post-hoc traces after harm. Observation without commit control is audit trail, not architecture. ControlPlane records entrance and interlocks commit; same graph yields exact dead compute.

2. **vs LLM-as-judge / static guardrails** — *does this look right?* vs *which span proves it?* Decision time = pure rule engine. Entitlement = identity, not classification. Default UNSUPPORTED is posture, not threshold tweak.

3. **vs RAG groundedness** — closest cousin, still short: retrieval-only, averages, action-blind. ControlPlane binds full provenance, prices **worst claim per action**, same unsupported claim → R1 Edit vs R3 Escalate. **Retrieval ≠ permission. Proof scales with consequence.**

Closer: Competitors publish precision. ControlPlane publishes misses and **refuses** to claim it eliminates hallucination/bias/privacy. Almost none disclaim themselves. Plane audited by the standard it enforces.

12. Fidelity + Content Laws

Invariant	Status
Default = UNSUPPORTED	Untouched
Entitlement / ACL set-membership; zero LLM	Untouched
Exact R×S; no route parameter	Untouched
Hard gate on actions; speculative release forbidden	Untouched
Dual-action R1 Edit + R3 Escalate **held** (never “blocked”)	Untouched
FNR typed format; empty until earned	Untouched
`UNKNOWN` never → `SUPPORTED`	Untouched
No LLM-as-judge / confidence as primary path	Untouched
Bias = async route-level only	Untouched
Refuse-to-claim (about *us*)	Untouched
Exactly two Stage 1 live routes	Untouched
Latency ≤40 ms p50 / ≤200 ms p95	Untouched
Surgical edit · evidence packet · Lane 1 always on · locked R3	Untouched

Law	Exact rule
Clause 7.2	Does **not** exist. Absence ≠ conflict. Never “caps/denies/doesn’t cover.” → Unsupported+categorical → **Escalate**, not Block.
Held ≠ blocked	Refund **held and escalated with the evidence packet** — never “blocked.”
Who pays	**Company wrongly pays out.** Customer did not lose money.
Dual action	Text → R1×entitlement→Edit (C3). Refund → R3×unsupported-categorical→Escalate (C2). Both simultaneous.
Latency	≤40 ms p50 / ≤200 ms p95. Never quote 40 as p95. Speculative verify OK; speculative release forbidden. Hold-back ~150–300 ms.
Refuse-to-claim	No: eliminate hallucinations · zero integration · zero added latency · one accuracy number.

Closing spine: AI can act → unproven claim must not authorize → provenance outside model → Default UNSUPPORTED → entitlement set-membership → R×S prices by consequence → hard gate on commit → publish misses.

Vocabulary: authorise · admit · prove · bind · refuse · hold · escalate · gate — not monitor · detect · observe · watch · guard · trust score · risk score · “responsible AI” as virtue.

Once you accept that an AI response is a set of claims requesting permission to act, you must capture provenance outside the model, invert the burden of proof, carry identity into verification, and gate the commit *path. Any softer design is a different product.*

Appendix — Official brief coverage

Brief requirement (`docs/ps.md`)	Where	Freeze stance
Different risk/latency by use case	§5 RoutePolicy + profiles	Same graph/matrix; config only
Overlapping bias/hallucination/privacy	§§1, 5	Not one classifier
No reliable real-time ground truth	§§1–2; FNR null	Honesty over bluff
Over-/under-flag tradeoff	§§5, 8–10	Shadow earn-out; blast-radius pricing
Multi-turn / agents that act	§§1, 5 session ledger	No separate product
Evolving geo/industry regulation	§5 additive overlays	Cannot loosen matrix
API-only foundation models	§§2, 6	No weights/logits
Detection options (AI-as-judge, confidence)	§§2, 11	**Reject** as primary
Decision logic allow/edit/flag/block	§3 matrix	Transcribed
Architecture placement	§§2, 5–6	Pre-commit interlock + hold-back
Governance / audit	§5	Zero LLM at decision time
Feedback loops	§5	Calibration, not trained model
Metrics / FP/FN / trustworthiness	§§5, 8F	Publish misses as format
Multi use-case, tens of thousands/week, mixed governance	§§4–5	Directional; prototype simulated
Deliverables: proposal + prototype + pitch	Header map	Proposal=this · Prototype=`controlplane/` · Pitch=next

End of ControlPlane.ai Round 2 Dense Final Hybrid — Stage Check PASS. Pitch from this file + live demo.